



MASH

MISHMASH OPENING CONFERENCE · MACHINES SESSION

Building a Norwegian Ecosystem for AI and Creativity

Kilden, Kristiansand · 14 September 2026





THREE QUESTIONS I GET

1 "How do I get access to train a model on Sigma2?"

Which tools *can* we use?

2 "Can I get funding for Claude Code?"

Which tools *should* we (not) use?

3 "How do I send an avatar of myself from Bergen to Trondheim in 10 milliseconds?"

Which tools do we need to *develop*?





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Rent or run





Rent a service

ChatGPT · Claude · Gemini · Suno · Runway · Midjourney

- + the best models, today
- + zero setup, pay per use
- + works on any device

- no weights, no reproducibility
- your data leave the room
- terms and prices change
- same for everyone, no yours

control: theirs

The middle road

open models,
rented GPUs
(HF, Mistral,
EU providers)

Run it yourself

open models on hardware you or your institution control

- + full control, reproducible
- + private: nothing leaves
- + can be trained on your material

- one step behind the frontier
- you carry the setup
- you pay the energy bill
- needs hardware and time

control: yours





Six questions to ask of any tool

Which model?

Name and version. "AI" is not an answer.

Where does it run?

Your machine, your institution, an EU data centre, the US.

Where do my data go?

Prompt, files, recordings. Stored? Trained on?

Who owns the output?

And can you sell a work made with it?

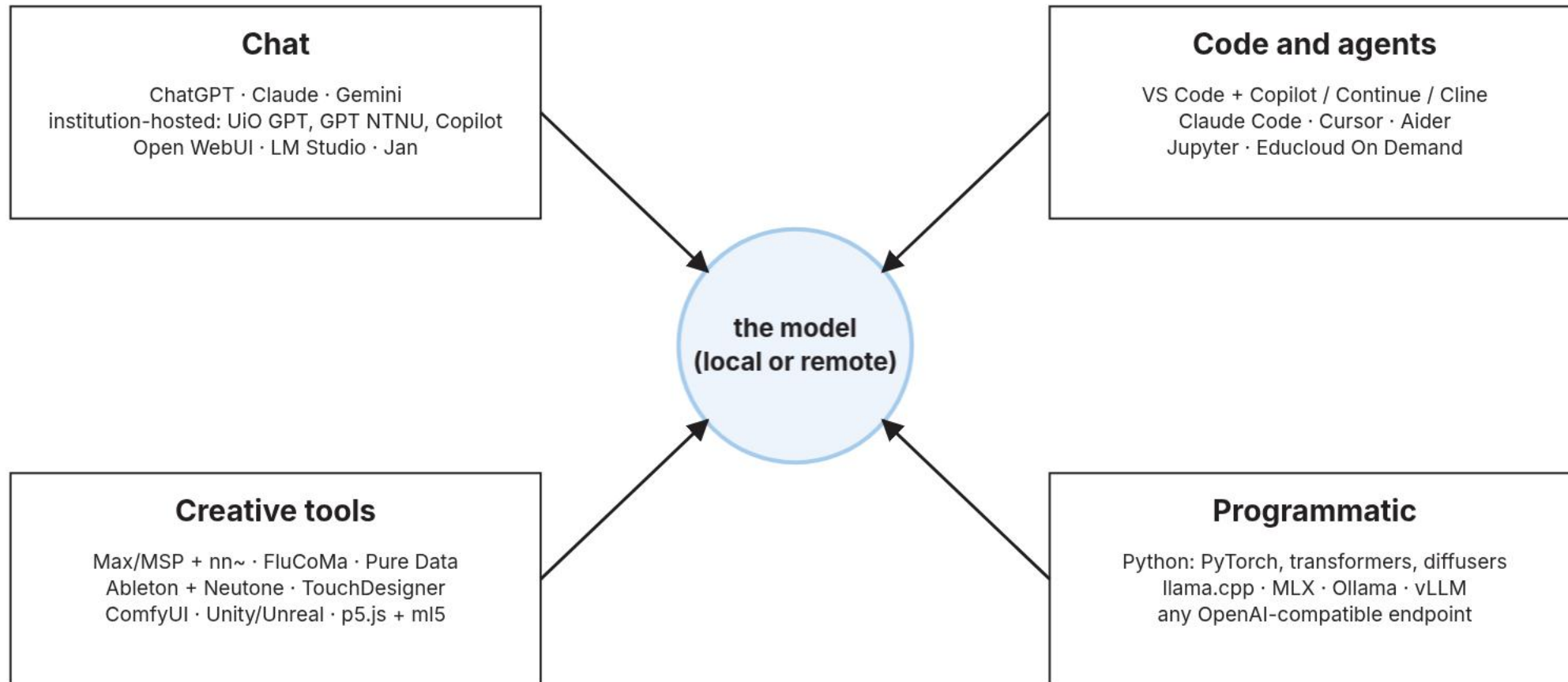
Same result next year?

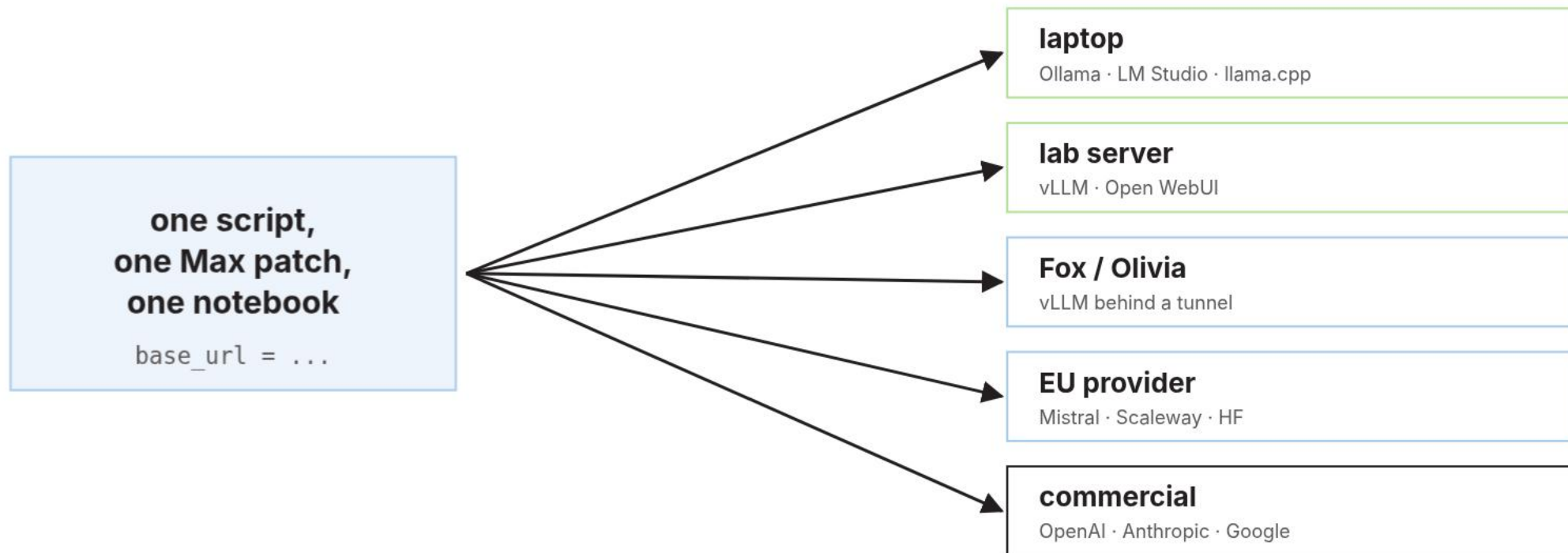
Reproducibility for research; continuity for a repertoire piece.

What does it cost?

Per month, per hour of GPU, per kilowatt-hour.



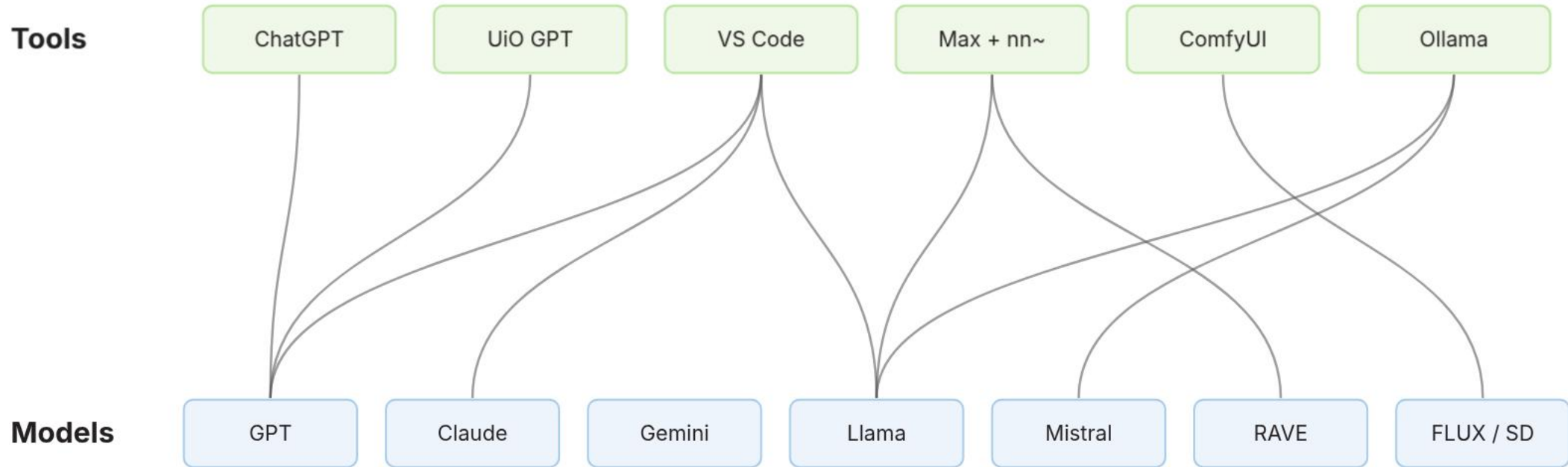




the same OpenAI-compatible API on all of them: change one line, keep the workflow



Tool is not model





Language

Llama · Mistral · Gemma · Qwen ·
DeepSeek · Phi

Norwegian: NorMistral (UiO) · NorLLM
(NorwAI/NTNU) · NB-GPT

Speech

Whisper · faster-whisper · Piper · XTTS
· Kokoro

Norwegian: NB-Whisper (National
Library)

Music and audio

RAVE (IRCAM) · MusicGen / AudioCraft
· Stable Audio Open · ACE-Step · YuE

Norwegian: none

Image

Stable Diffusion · SDXL · FLUX

Norwegian: none

Video

Wan · HunyuanVideo · LTX

Norwegian: none

Vision, motion, multimodal

CLIP · SAM · LLaVA · Qwen-VL · MDM ·
MotionGPT

Norwegian: none





Licences in three colours

Anything goes

Apache 2.0 · MIT

- Mistral, Mixtral · Qwen3 · DeepSeek · Phi
- NorMistral (UiO) · NB-Whisper
- Whisper · RAVE · CLIP · SAM
- FLUX.1 schnell · Wan 2.1 · YuE · ACE-Step
- use it, sell the work, ship the product

Conditions

Community licences and terms of use

- Llama: "Built with Llama", 700M-user cap, derivatives named Llama
- Gemma: prohibited-use policy passed on
- Stability community licence (SD3.5, Stable Audio Open): free under \$1M revenue
- CreativeML OpenRAIL-M (SD 1.5, SDXL): use restrictions travel with the weights
- NorwAI NorLLM: Nordic organisations

Non-commercial

Research only, CC-BY-NC

- **FLUX.1 dev: non-commercial licence for the model**
- **MusicGen: code MIT, weights CC-BY-NC 4.0**
- **many research checkpoints on Hugging Face**
- **fine for a paper; a commissioned work is commercial**



Licence compatibility: what you can do with what

Base model licence	Research paper	Commissioned work, sold	Product or service	Release your fine-tune	Mix with a permissive model	Train a new model on its outputs
Apache 2.0 / MIT Mistral, Qwen3, NorMistral, Whisper, RAVE, FLUX schnell	•	•	•	• any licence, keep the notice	•	•
Llama community Llama 3, Llama 4	•	• "Built with Llama"	🔒 under 700M users	🔒 same licence, name starts with Llama	🔒 the mix inherits Llama terms	🔒 allowed, new model named Llama
Gemma terms of use Gemma 2, 3	•	•	🔒 prohibited-use policy	🔒 terms passed on	🔒 the mix inherits the terms	🔒 check the terms
Stability community SD3.5, Stable Audio Open	•	🔒 free under \$1M revenue	🔒 same cap, else enterprise licence	🔒 same licence	🔒 the mix inherits the cap	🔒 check the terms
OpenRAIL-M SD 1.5, SDXL	•	•	🔒 use restrictions apply	🔒 restrictions passed on	🔒 restrictions travel with it	🔒
Non-commercial FLUX.1 dev, MusicGen weights, CC-BY-NC	•	🔒 read it: outputs sometimes yes, the model no	○	🔒 non-commercial only	○ the whole release becomes non-commercial	○ usually forbidden





Where to find them, how to trust them

Find

- Hugging Face: weights, model cards, datasets
- Ollama and LM Studio libraries: one-click local
- IRCAM Forum: RAVE and nn~
- the paper's GitHub, for the newest

Trust

- model card: training data disclosed?
- evaluation: on what, by whom?
- reproducible: pinned version, checksum
- energy: training and inference cost stated?



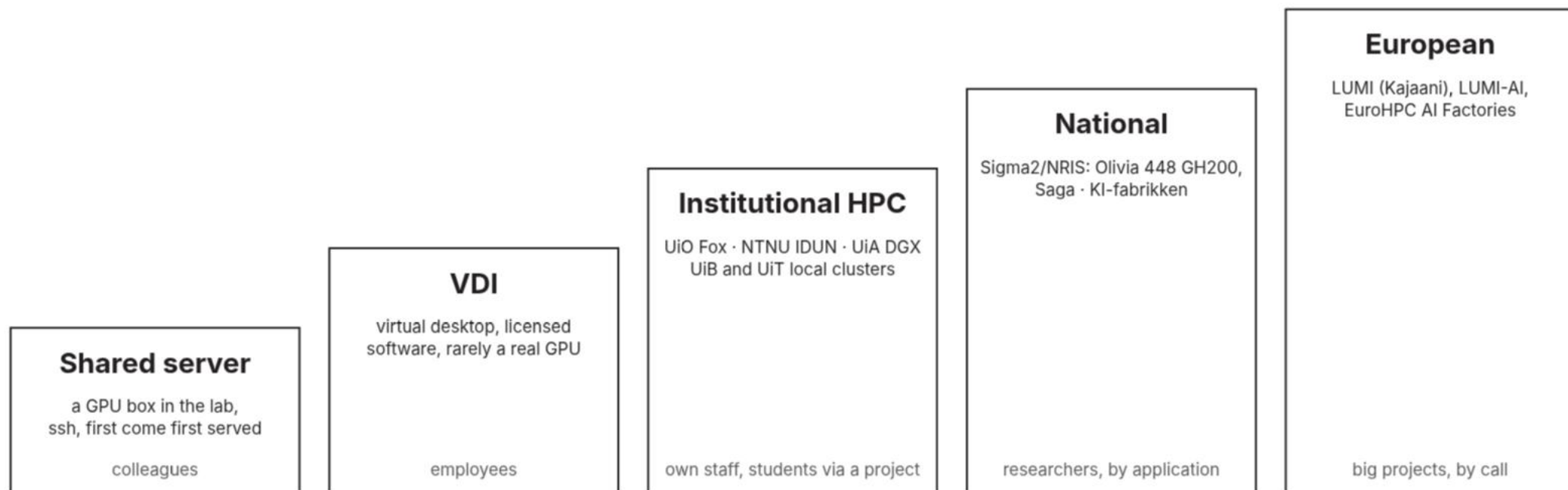


Where the compute is



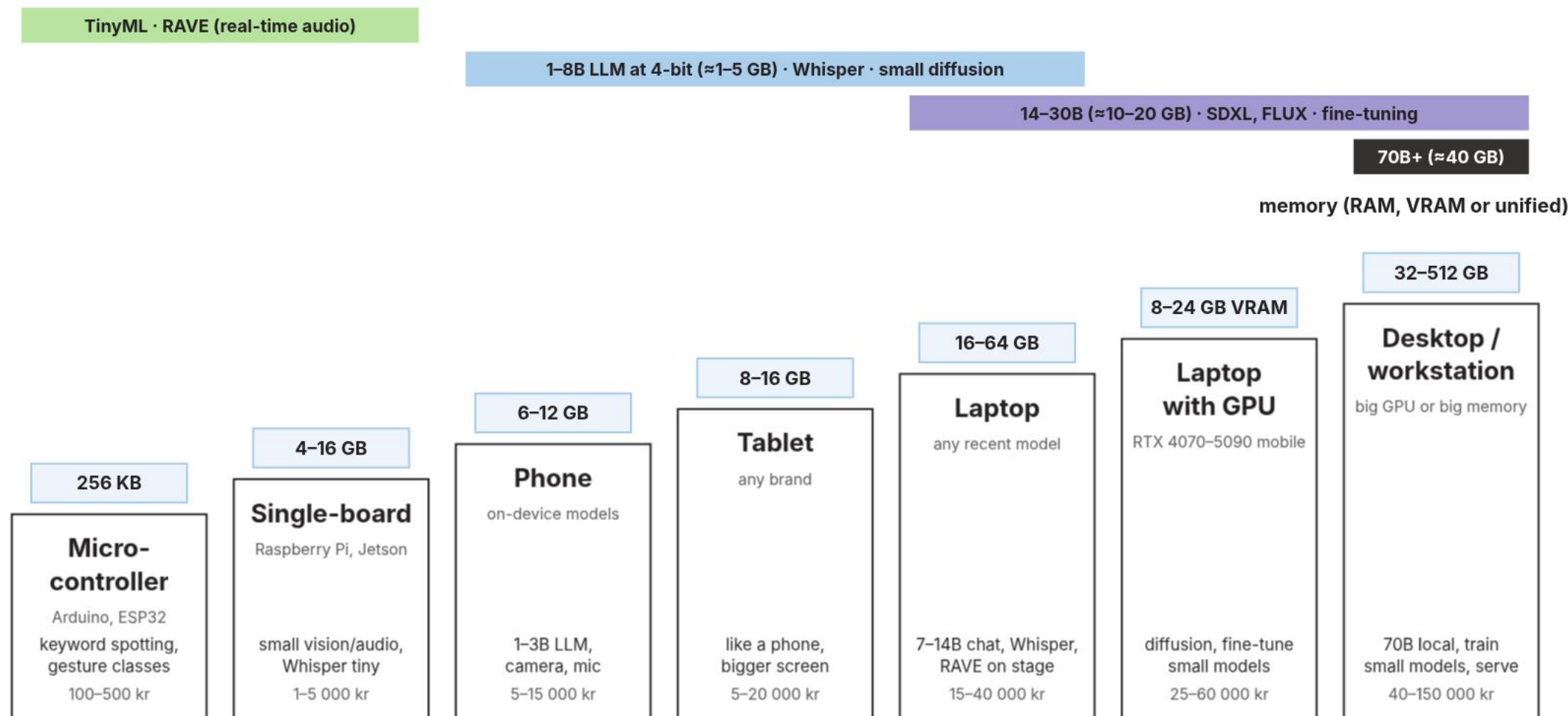


more compute · more people can share it · more data next to it



more waiting · more paperwork · less interactive · batch, not a rehearsal







parameters × bytes per parameter ≈ memory

7–8B	4-bit	≈ 5 GB	any recent laptop or phone
30B	4-bit	≈ 20 GB	laptop with 32 GB unified, or a 24 GB GPU
70B	4-bit	≈ 40 GB	large unified memory, or two big GPUs
70B	16-bit	≈ 140 GB	a server, or Fox



What each rung is good for

Microcontroller

- keyword spotting, gesture classes
- sensor fusion, milliseconds
- no LLM, ever
- wearables, instruments

Pi / Jetson

- small vision and audio models
- Whisper tiny, YOLO
- an embodied agent in a box
- installations that run for a year

Phone / tablet

- 1–3B on-device LLMs
- camera, mic, sensors
- closed platforms, app stores
- the device every pupil has

Laptop

- 7–14B chat, offline
- Whisper transcription
- RAVE and nn~ on stage
- the studio in a bag

Laptop with GPU

- Stable Diffusion, FLUX
- fine-tune small models (LoRA)
- real-time everything
- hot, loud, one hour of battery

Workstation

- 70B local, video generation
- train RAVE overnight
- serve a whole studio or class
- needs a room and a power bill

School tablet or laptop

- managed device, no GPU
- Feide login, pupils' data
- everything must be hosted
- the most constrained rung

Rule of thumb

- memory decides what fits
- compute decides how fast
- prototype on the smallest rung that works





How you actually get on

Institutional clusters

- UiO Fox: Educloud project, PI adds collaborators
- NTNU IDUN: supervisor requests access
- UiA: DGX servers at CAIR
- UiB and UiT: local HPC via the IT division

NRIS / Olivia

- project application, CPU and GPU hours
- two rounds a year
- any Norwegian research institution
- data storage next to compute

KI-fabrikken

- Sigma2, part of the LUMI AI Factory network
- research, business, public sector
- expertise and support, not only GPUs
- artists and cultural institutions: unclear

LUMI / EuroHPC

- through NRIS or EuroHPC calls
- AMD GPUs: check your software
- large allocations, long lead times

NAIC (paused)

- the low-threshold national GPU service
- funding ended 31 March 2026
- "no-cost hibernation" since April
- successor being procured by Sigma2

Sensitive data

- TSD at UiO, SAFE at UiB, HUNT Cloud at NTNU
- GPU capacity inside is limited
- health and pupils' data live here or offline





Institutions side by side · to be filled in

Institution	Own GPU cluster	Hosted chat / LLM service	Sensitive-data environment	Students get GPUs?	Freelancers / partners get in?
UiO	Fox	UiO GPT	TSD	via project	via a PI
UiB	local HPC (UiB IT)	Copilot	SAFE	?	?
NTNU	IDUN	GPT NTNU	HUNT Cloud	via supervisor	support agreement needed
UiT	local HPC group; hosts NRIS systems	?	?	?	?
UiA	DGX-2 and DGX H100 (CAIR)	?	?	?	?
INN · HiØ · HVL · OsloMet · Nord · Kristiania ...	?	?	?	?	?
NMH · KHiO · AHO	none known	?	none	no	?
National Library	DH-lab	?	?	n/a	?
Simula · SINTEF · IFE	eX3 and own	?	?	n/a	?





Inference is not training

Serving a model

- always on, answers in milliseconds
- a web endpoint that a class, a stage or a museum connects to
- a modest GPU, or many of them, or a laptop
- needs networking, logins and a security review

Training and fine-tuning

- bursts: hours to weeks, then nothing
- no endpoint, no users, a queue
- the biggest GPUs and the fastest storage
- needs scheduling, checkpoints, data next to compute



Workflows do not travel

Laptop to Fox

Different modules, versions, paths. Days of setup.

Fox to Olivia

Same scheduler, different drivers, containers and rules.

Olivia to LUMI

NVIDIA to AMD, CUDA to ROCm. Often a rewrite.

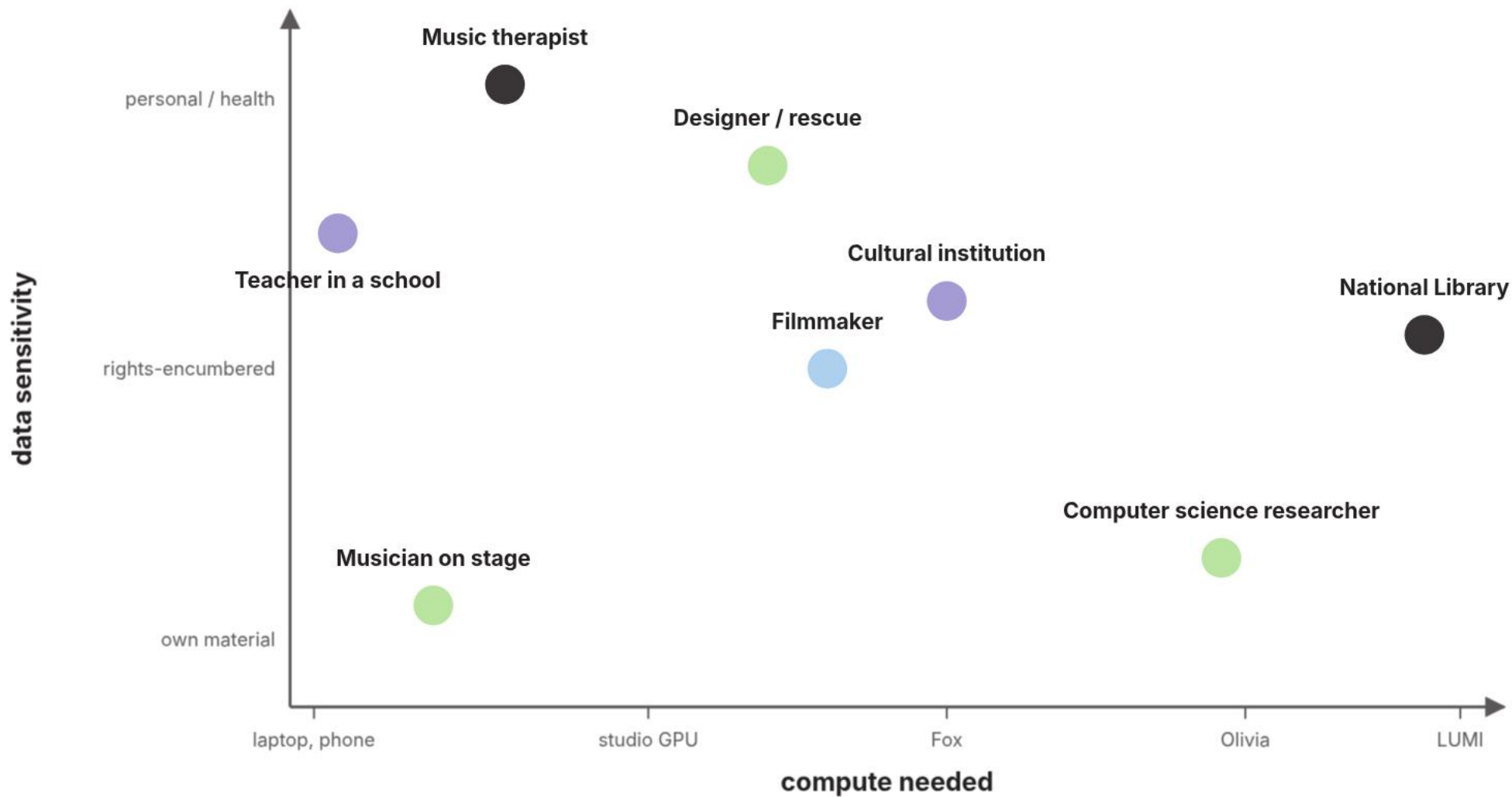
What helps

Containers and pinned environments. Teach it at student level. AI assistants make porting cheap.

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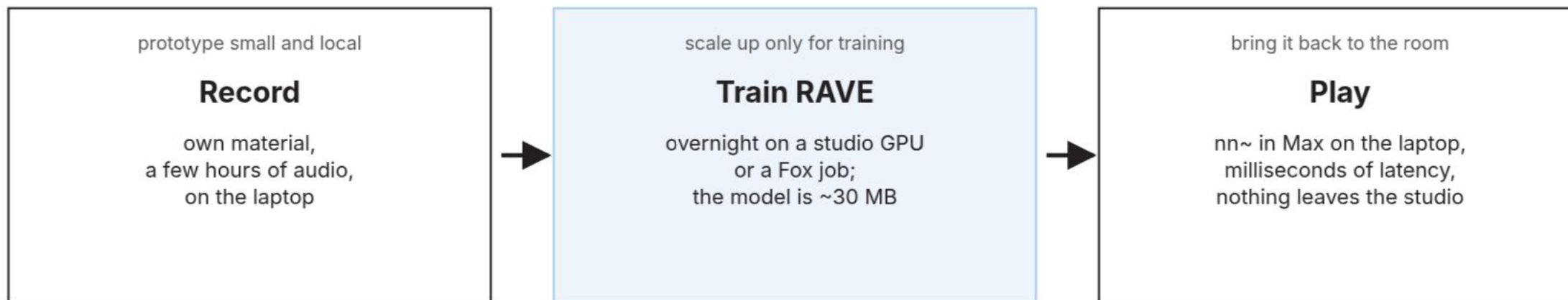
Who needs what





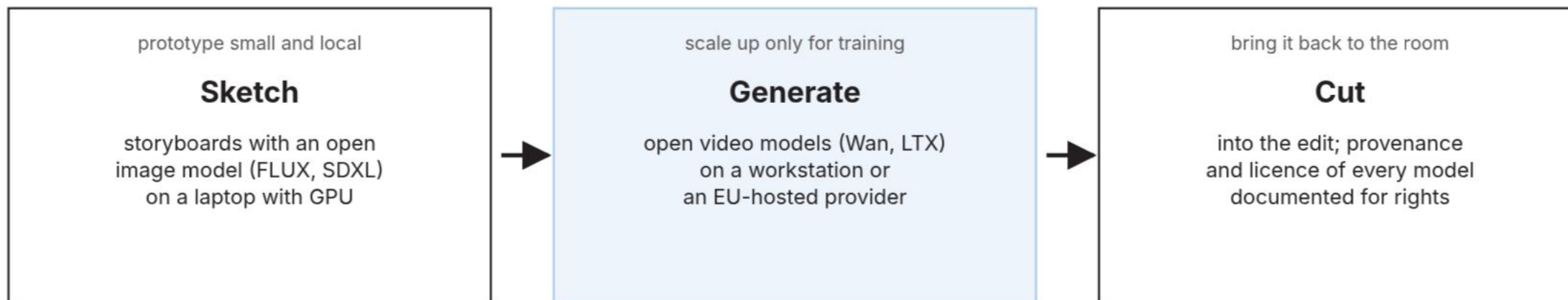


WP1 · A musician on stage





WP2 · A filmmaker





WP4 · A teacher in a school



WP6 · The National Library

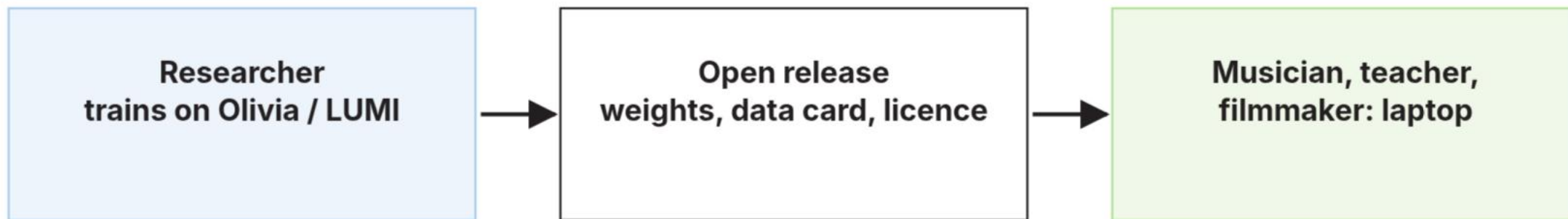




Work Package Needs

WP	Typical machine task	Data	Compute	Runs today on	The hole
1 Performances	real-time co-creative agents, RAVE, gesture models	own recordings, sensors	ms; laptop, board, one GPU to train	laptop, studio GPU, Fox	open real-time models; training time without paperwork
2 Processes	image, video, text, audio generation	rights-encumbered	hours on a workstation	commercial tools, workstation	rights-clean open models; EU hosting
3 Health	small models on sessions and signals	health, personal	small, inside a secure zone	TSD, offline laptops	GPUs inside secure zones; consent-ready pipelines
4 Education	AI literacy, classroom tools	pupils' data	none locally; hosted	tablets and laptops, Feide, hosted chat	free, Norwegian, GDPR-safe creative tools
5 Industries	production generation, attribution, audits	contracts, identity	hosted, reliable, auditable	commercial APIs	cultural institutions in KI-fabrikken; open-provider contracts
6 Heritage	transcribe, classify, link, train on the collection	petabytes; copyright, public duty	national or European HPC	NB DH-lab, Olivia, LUMI	multimodal Norwegian models; sustained allocations; storage next to compute
7 Problem-solving	constrained generation, embodied systems	industrial, safety-critical	edge to workstation; simulation	partner systems, Fox	constraint-aware models; access for industry partners





when the loop works: compute upstream, open models downstream





Develop it yourself

two different jobs

Use off the shelf

finished model



any device

hardware: anything that runs inference
who: everyone on the map

time  minutes

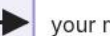
the catch: their data, their terms

Train an existing architecture on new data

your data



fixed architecture



your model

RAVE on your recordings, a LoRA on your film
hardware: one GPU, sometimes a cluster
who: PhDs, labs, a studio with a GPU

time  hours to days, known

the catch: rights to the data

Develop new methods



new architectures, objectives, control
hardware: many runs, most fail
who: ML groups on national compute

time  ?

the catch: allocations want a number first



CREATIVE AI IS NOT "TRADITIONAL" AI

AI for a task

- one task, one right answer
- batch: throughput matters, latency does not
- large, clean, labelled data
- measured by accuracy on a benchmark
- runs in a data centre, out of sight

AI for creative work

- open-ended: many good answers, or none
- real-time and interactive: latency is the constraint
- small, personal, messy data, often your own
- measured by taste, context and who is asking
- on stage, in the room, on the body: control matters



Tasks against platforms: a rough guide

Task	Commercial API	Laptop	Laptop + GPU	Workstation	Fox-type cluster	Olivia / LUMI
Chat, writing, code	•	• 7-14B	•	• 70B	🕒 overkill	🕒 overkill
Transcription (Whisper)	•	•	•	•	🕒 batch	🕒 overkill
Image generation	•	🕒 slow	•	•	🕒 batch	🕒 overkill
Video generation	•	○	🕒 short clips	•	🕒 batch	🕒
Real-time audio on stage	○ latency	•	•	•	○	○
Train a small model (RAVE, classifier)	○	🕒 slow	•	•	•	🕒 overkill
Fine-tune an LLM (7-14B)	🕒 data leave	○	🕒 LoRA	•	•	•
Train from scratch (NB-scale multimodal)	○	○	○	○	🕒 small	•
Sensitive data (health, pupils)	○	• offline	• offline	• offline	🕒 TSD-type only	○



By person: a plausible stack today

Who	Everyday	Creative work	Training	The catch
Musician on stage	laptop, local chat	Max + nn~, RAVE	studio GPU overnight, or a friend with Fox	no institution: no Fox, no NRIS
Filmmaker	commercial tools	ComfyUI, open video models on a workstation	LoRA on own footage	rights of the base model
Teacher in a school	Feide-gated hosted chat	browser tools only	none	no Norwegian, GDPR-safe creative tool exists
Music therapist	offline laptop	small local models	inside TSD, if GPUs	consent and secure-zone capacity
Bachelor student	Ollama + Open WebUI	Whisper, notebooks	Colab-type, or a course allocation	depends entirely on the institution
PhD / ML researcher	laptop	prototypes locally	Fox, then NRIS, then LUMI	allocation rounds and the paperwork
Cultural institution	commercial APIs	hosted open models with a contract	KI-fabrikken, if they qualify	nobody has defined them as a user category
Startup / studio	commercial APIs	own GPUs or cloud	KI-fabrikken as "business"	cost, and no research partner
National Library	own lab	own models	Olivia, LUMI	sustained multi-year allocation





Holes in the map





Who	Own hardware	Institutional servers, VDI, Fox-type	National HPC NRIS, Olivia	KI-fabrikken	Hosted open models EU providers	Commercial APIs
Researchers, big universities UiO · UiB · NTNU · UiT	yes	yes, varies	by application	yes	own budget	own budget
Researchers, other universities and colleges UiA · INN · HiØ · HVL · NMH · KHIO · AHO · Kristiania ...	yes	often little or none	by application	yes	own budget	own budget
Research institutes	yes	own labs	by application	yes	own budget	own budget
Students, bachelor to PhD	own machine	via a supervisor	via a project	?	rarely	own budget
Teachers and schools	tablets, laptops, no GPU	no	no	no	no	Feide-gated, limited
Freelance artists	if affordable	no	no	?	own budget	own budget
Public sector museums · libraries · NRK · municipalities	yes	own IT, no GPUs	no	"public sector"	budget, procurement	budget, procurement
Private sector studios · film · games · startups	yes	own	only with a research partner	"business"	budget	yes



Data: open by count, closed by volume

Open, with an API

- Wikidata, KulturNav, the authority files
- half of DigitaltMuseum, Nasjonalmuseet
- Kartverket, SSB, Stortinget, Brønnøysund
- Språkbanken: thousands of hours of speech

Closed, by the hour and the page

- the library's books, newspapers, radio, TV, web
- NRK, TV 2, NTB, Retriever
- Bokhylla, Filmarkivet
- nothing open for music, film, video, dance, theatre, games, design



Observations to argue with

- The academic infrastructure is built for batch science, not for a rehearsal room.
- The people who most need cheap, controllable, rights-clean models have the least access.
- Access depends on which institution you sit in, not on what you need.
- Open Norwegian models exist for language and speech, and for nothing creative.
- The whole Norwegian cultural heritage sits in one collection. Nobody has trained a multimodal model on it.
- Nobody has defined "artist", "teacher" or "cultural institution" as a user of national compute.
- The low-threshold national service (NAIC) is paused; its successor is not yet here.
- Workflows are locked to the system they were built on. Fox is not Olivia, and Olivia is not LUMI.
- When big systems are bought, the users are rarely asked first.
- Norway's data is open by count and closed by volume. The library is licensed for reading, not mining.
- No text-and-data-mining exception in law. The bill is in committee; input is due 28 September.
- 200 researchers on a top-tier subscription is 6 million kroner a year. Per seat is the wrong shape; per use through a gateway is the right one.
- There will never be unlimited access to the best models. Good enough, on a quota, is the plan.
- One PhD fellowship is 150 seat-years. Tokens multiply people; they do not replace them.

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The bill





ONE SUBSCRIPTION · 200 RESEARCHERS

6 million kroner a year

200 researchers on one top-tier agent subscription at 2 500 kroner a month.



Two ways out, and the elephant

National: quotas through one door

- the gateways and clusters of the big institutions, opened to everyone else on the map
- a quota per person, paid per use by the unit, across vendors
- frontier models when they are needed, open models for the rest
- quota categories for artists, schools and museums, not only researchers

Local: small and smart

- small open models on the laptop, the board, the phone
- several small agents, one job each, instead of one large model for everything
- energy and cost in the room, where you can see them
- for the stage, the classroom, the museum

The elephant: the bill that matters is not 200 researchers. It is generated feeds and the rest of the consumption nobody asked for. I would argue a centre for AI and creativity has to confront that, and show what else is possible.





Tokens or people

What	Kroner	The same money as
One seat, one year	30 000	1 seat-year
One PhD year	1 400 000	47 seat-years
200 seats, one year	6 000 000	1.3 fellowships, or 4 PhD years
200 seats, five years	30 000 000	7 fellowships





Pay per use, not per seat

One door

- an institution-hosted gateway in front of commercial and open models: UiO GPT, GPT NTNU
- chat, shared assistants, files, and now API keys for agents and coding tools
- about 50 public-sector institutions run it

Keys by data class

- one personal key per data class: red, yellow, green
- the key is independent of the model
- a monthly quota per model
- a unit authorises a quota and pays for actual use

Good enough, not best

- the best open-weight models, served on Fox and IDUN, cover most jobs
- smaller models win on price per result
- the frontier models on a quota when they are needed, without a seat per person



A PROPOSAL TO ARGUE WITH

How a centre could decide

A share, not a line

- budget tokens as a share of personnel cost, as equipment always was
- 3 to 5 per cent of a PhD year is 40 000 to 70 000 kroner
- enough for heavy pay-per-use through a gateway

Quotas by role

- a PhD in machine learning, a WP leader and a museum partner do not need the same budget
- the gateway model, applied to the centre's own money
- the unit sees what its people actually use

Not fungible

- fellowships are in the contract; tokens come from operating money
- so the real trade-off is tokens against a lab engineer, workshops, travel, hardware in the room
- 50 seats is one engineer who builds the local alternative

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Over to the panel



- 
- 1 Which tools *can* we use?
 - 2 Which tools *should* we (not) use?
 - 3 Which tools do we need to *develop*?

Stefano Fasciani

University of Oslo,
Department of Musicology

Enrique Encinas

Oslo School of Architecture
and Design

Anna-Maria Christodoulou

University of Oslo, RITMO

Keith Mellingen

VRINN, immersive learning
cluster, Hamar



What do you use, and what can you not get?

nettskjema.no/a/649352

